

PATIENT HISTORY:

CHIEF COMPLAINT/ PRE-OP/ POST-OP DIAGNOSIS: Thyroid carcinoma.
PROCEDURE: Total thyroidectomy/right neck dissection.
SPECIFIC CLINICAL QUESTION: Not given.
OUTSIDE TISSUE DIAGNOSIS: Not given.
PRIOR MALIGNANCY: Not given.
CHEMORADIATION THERAPY: Not given.
ORGAN TRANSPLANT: Not given.
IMMUNOSUPPRESSION: Not given.
OTHER DISEASES: Not given.

UUID: 026940CA-36B1-4AF8-996D-18F8D108CD7A
TCGA-BJ-A0ZB-01A-PR

Redacted



10D-0-3

ADDENDA:**Addendum**

carcinoma, papillary, tall cell variant
8344/3

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Site: thyroid, NOS c73.9

Block 2B:

hw 11/27/11

- A. **BRAF V600E mutation IDENTIFIED.**
- B. **Mutations in NRAS61, HRAS61, KRAS12/13 NOT identified.**
- C. **FISH test results are negative for RET/PTC rearrangement.**

NOTE:

DNA was extracted in the amount sufficient for testing.

BACKGROUND:

Mutations in either *BRAF* or *RAS* genes or *RET/PTC* rearrangements are found in more than 70% of papillary thyroid carcinomas (1). *BRAF* V600E (T1799A) mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between *BRAF* V600E mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the *RAS* genes or *PAX8/PPARg* rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocyctic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, *BRAF* V600E mutation and *RET/PTC* and *PAX8/PPARg* rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). *RAS* mutations occur in malignant and benign thyroid tumors, being found in ~40-50% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

1. Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. *Am J Surg Pathol* 2006, 30:216-222
2. Xing M, et al. *BRAF* mutation predicts a poorer clinical prognosis for papillary thyroid cancer. *J Clin Endocrinol Metab* 2005, 90:6373-6379.
3. Elisei R, et al. *BRAF* V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. *J Clin Endocrinol Metab* 2008, 93:3943-3949.
4. Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential *BRAF*(V600E) mutational status in high aggressive papillary thyroid microcarcinoma. *Ann Surg Oncol*. 2009 Feb;16(2):240-5.
5. Nikiforova MN, et al. *RAS* point mutations and *PAX8-PPARg* rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. *J Clin Endocrinol Metab* 2003, 88: 2318-26.
6. Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV., ed. *Endocrine Pathology: Differential Diagnosis and Molecular Advances*. Humana Press, Totowa, 2004, 191-209.

MUTATIONAL ANALYSIS:

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the LightCycler platform to amplify *BRAF* codons 599-601, *NRAS* codon 61, *HRAS* codon 61, and *KRAS* codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. If required, the mutation type was confirmed by Sanger sequencing of the PCR product on DNA from samples positive for each of these mutations was used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal DNA.

Addendum

PROBE: **MEC 1 Dual Color Break-apart Probe**
Cytogenetic Location: 11q21

MEC1 FISH STUDIES ARE NEGATIVE FOR TRANSLOCATION.

cells analyzed: 60

cells with translocation pattern: 0

cells with the normal pattern: 60(100%)

MAML2 translocation detection is as follows: Normal cells without the *MAML2* translocation contain two fusion signals. Cells with the translocation pattern show one fused signal (green and orange fusion) and one separate green and one separate orange signal. Targeted areas containing greater than 20% of cells with the translocation pattern are considered positive.

MEC1 FISH analysis was manually performed and quantitatively assessed by analysis of a minimum of 60 cells using the *MAML2* SpectrumOrange (telomeric) and the *MAML2* SpectrumGreen (centromeric) probes.

Criteria	Yes	No
Diagnosis Discrepancy		☒
Primary Tumor Site Discrepancy		☒
HIPAA Discrepancy		☒
Prior Malignancy History		☒
Dual/Synchronous Primary		☒
Case is (circle):	QUALIFIED	DISQUALIFIED
Reviewer Initials	hw	11/27/11

References:

Behboudi A, Enlund F, Winnes M, Andrén Y, Nordkvist A, Leivo I, Flaberg E, Szekely L, Mäkitie A, Grenman R, Mark J, Stenman G. *Molecular classification of mucoepidermoid carcinomas-prognostic significance of the MECT1-MAML2 fusion oncogene.* Genes Chromosomes Cancer. 2006 May;45(5):470-81.

Okabe M, Miyabe S, Nagatsuka H, Terada A, Hanai N, Yokoi M, Shimozato K, Eimoto T, Nakamura S, Nagai N, Hasegawa Y, Inagaki H. *MECT1-MAML2 fusion transcript defines a favorable subset of mucoepidermoid carcinoma.* Clin Cancer Res. 2006 Jul 1;12(13):3902-7.

Tonon G, Modi S, Wu L, Kubo A, Coxon AB, Komiya T, O'Neil K, Stover K, El-Naggar A, Griffin JD, Kirsch IR, Kave FJ. *t(11;19)(q21;p13) translocation in mucoepidermoid carcinoma creates a novel fusion product that disrupts a Notch signaling pathway.* Nat Genet. 2003 Feb;33(2):208-13. Epub 2003 Jan 21. Erratum in: Nat Genet. 2003 Mar;33(3):430.

FINAL DIAGNOSIS:

PART 1: LYMPH NODE, CENTRAL COMPARTMENT, RIGHT SIDE, RIGHT NECK DISSECTION –

- A. METASTATIC PAPILLARY THYROID CARCINOMA, TALL CELL VARIANT IN FIVE OF SIX LYMPH NODES (5/6).
- B. EXTRANODAL SPREAD IS SEEN.
- C. VENOUS INVASION IS IDENTIFIED.

PART 2: THYROID, TOTAL THYROIDECTOMY (16.0 GRAMS) –

- A. PAPILLARY THYROID CARCINOMA, TALL CELL VARIANT, 2.4 CM, RIGHT LOBE (see comment).
- B. EXTRATHYROIDAL EXTENSION IS SEEN.
- C. THE TUMOR EXTENDS TO THE MARGIN.
- D. NODULAR THYROID WITH HYALINIZATION AND CALCIFICATION.
- E. NO PARATHYROID GLANDS ARE SEEN.
- F. PATHOLOGIC STAGE pT3, N1b, MX.

PART 3: LYMPH NODE, RIGHT LATERAL COMPARTMENT, RIGHT NECK DISSECTION –

- A. SOFT TISSUE (?LYMPH NODE) WITH METASTATIC PAPILLARY CARCINOMA, TALL CELL VARIANT WITH EXTENSIVE SQUAMOUS METAPLASIA.
- B. TEN BENIGN LYMPH NODES (0/10).

COMMENT:

This case has been reviewed by

r and all concur with the above diagnosis.

Molecular studies for thyroid carcinoma are being done and the results will follow in an addendum. We are also doing FISH study for MAML2 to further evaluate the areas of squamous metaplasia and, more specifically, to rule out a primary mucoepidermoid carcinoma of the thyroid.

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE:	Total Thyroidectomy
TUMOR SITE:	Right Lobe ✓
TUMOR FOCALITY:	Unifocal ✓
TUMOR SIZE (largest nodule):	Greatest Dimension: 2.4 cm Additional Dimensions: 1.6 X1.2 cm
HISTOLOGIC TYPE:	Papillary carcinoma, tall cell variant
PATHOLOGIC STAGING (pTNM):	pT3 pN1b Number of regional lymph nodes examined: 17 Number of regional lymph nodes involved: 6 pMX
MARGINS:	Margin(s) involved by carcinoma Site(s) of involvement: right lobe
VENOUS/LYMPHATIC (LARGE/SMALL VESSEL) INVASION (V/L):	Present
ADDITIONAL PATHOLOGIC FINDINGS:	Nodular goiter

SPECIAL PROCEDURES:

In Situ Procedure

Interpretation

RESULTS:

RET/PTC: The targeted area of the tissue showed 1(1.6%) of the cells with the rearrangement pattern and 60(98.4%) of the cells with the normal pattern. The targeted area is considered NOT rearranged for the *RET/PTC* region.

The *RET/PTC1* Probe did not contain the rearrangement pattern.

My signature is attestation that I have personally reviewed the submitted material(s) and the above diagnosis reflects that evaluation.

Results

RET/PTC Methodology Description:

Probe: *RET/PTC* Probe (10q11.2) and *RET/PTC1* Dual-Color Fusion Probe 10q11.2-10q21

Probe Description: The *RET* probe (labeled SpectrumGreen) is a 207 kb DNA probe located at band 10q11.2.

The H4 and 369 probes are approximately 100kb each (labeled in _____), and are located at band 10q21.

RET/PTC rearrangement is suggested to be present when three *RET* signals are identified in greater than 8% of the analyzed cells.

RET/PTC1 rearrangement detection is as follows: Normal cells without the *RET/PTC1* rearrangement show two pair of green and orange signals separated. Cells with the *RET/PTC1* rearrangement show two pair of signals (green and orange) in juxtaposition and one green and one orange signal separated in greater than 8% of analyzed cells within the targeted area.

RET/PTC FISH analysis was manually performed and quantitatively assessed by analysis of a minimum of 60 cells using the *PTC* (369/H4) SpectrumOrange and the *RET* probes.